UBNetDef S25 Instructional Report Homework #5

Adi Czobel

Edwin Kairu

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1. RockyDBServer: Configure database operations

```
MariaDB [(none)]> CREATE DATABASE mediawiki_db;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> use mediawiki_db;
Database changed
```

Figure 1: Create a database named mediawiki_db

In order to create a database named **mediawiki_db**, I used the command CREATE DATABASE mediawiki_db;

```
MariaDB [(none)]> CREATE USER 'sysadmin@10.43.10.20' IDENTIFIED BY 'Change.me!';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
```

Figure 2: Create a user called sysadmin

In order to create a user called **sysadmin**, I used the command: CREATE USER 'sysadmin'@'10.43.10.20' IDENTIFIED BY 'Change.me!';. In this context, **mediawiki_db** is the database name created with, **10.43.10.20** as the IP address of the MediaWiki server, and **Change.me!** is the password given to the user.

```
MariaDB [(none)]> GRANT ALL ON mediawiki_db.* TO 'sysadmin'@'10.43.10.20';
Query OK, 0 rows affected (0.001 sec)
```

Figure 3: Grant the New User Remote Privileges

In order to grant the new user remote privileges, the command GRANT ALL ON mediawiki_db.* TO 'sysadmin'@'10.43.c.20'**;** was executed. This command provides the user **sysadmin** with full access to all tables in the mediawiki_db database. The IP address **10.43.10.20** refers to the **RockyDBServer**, which is where the **mediawiki_db** database was created. This step allows the **sysadmin** user to connect to the database from the **RockyDBServer** and perform any necessary operations.

2. UbuntuWebServer: Configure MediaWiki

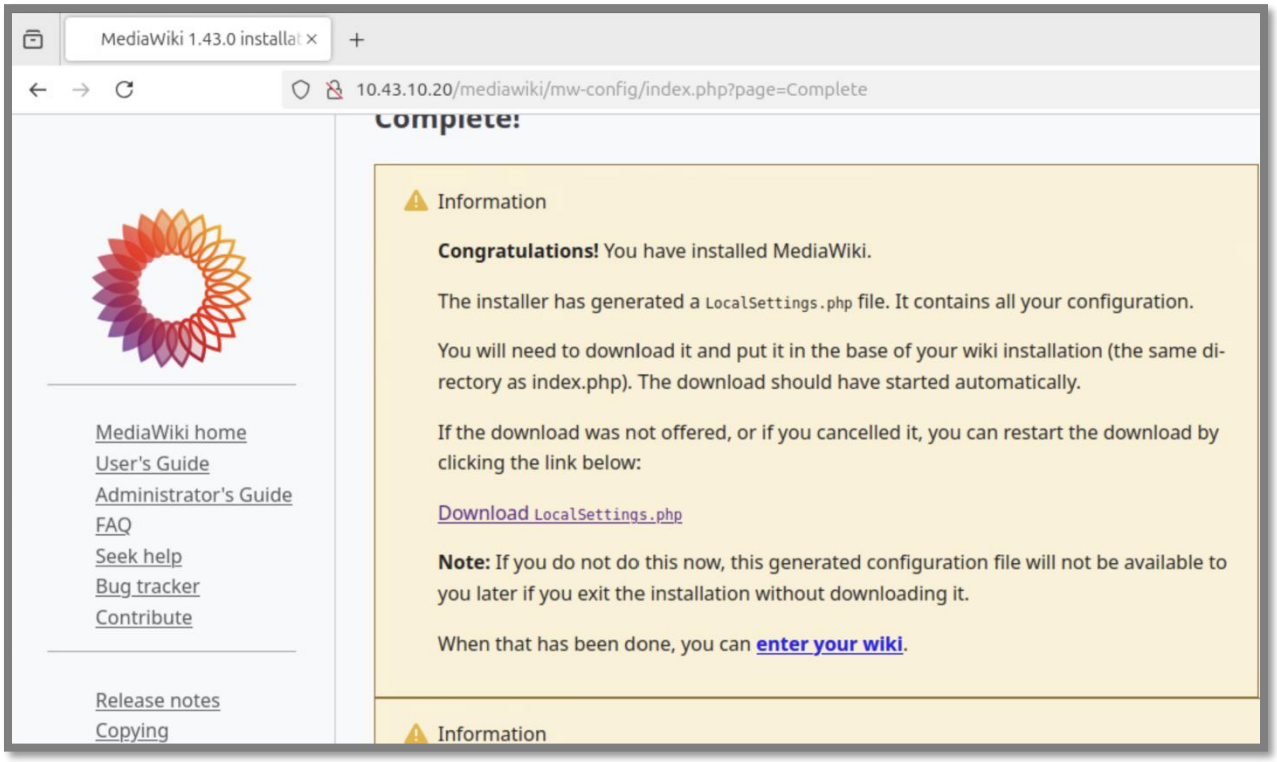


Figure 4: Initial Configuration Using the Provided ‘Wizard’ Feature

In the figure above it shows that MediaWiki has been successfully configured using the provided 'Wizard' feature. thus, the system is now ready, and the file **LocalSettings.php** is available for download. This file contains important configuration settings that need to be placed in the root directory of the MediaWiki installation for the site to function correctly.

3. Proof of Completion

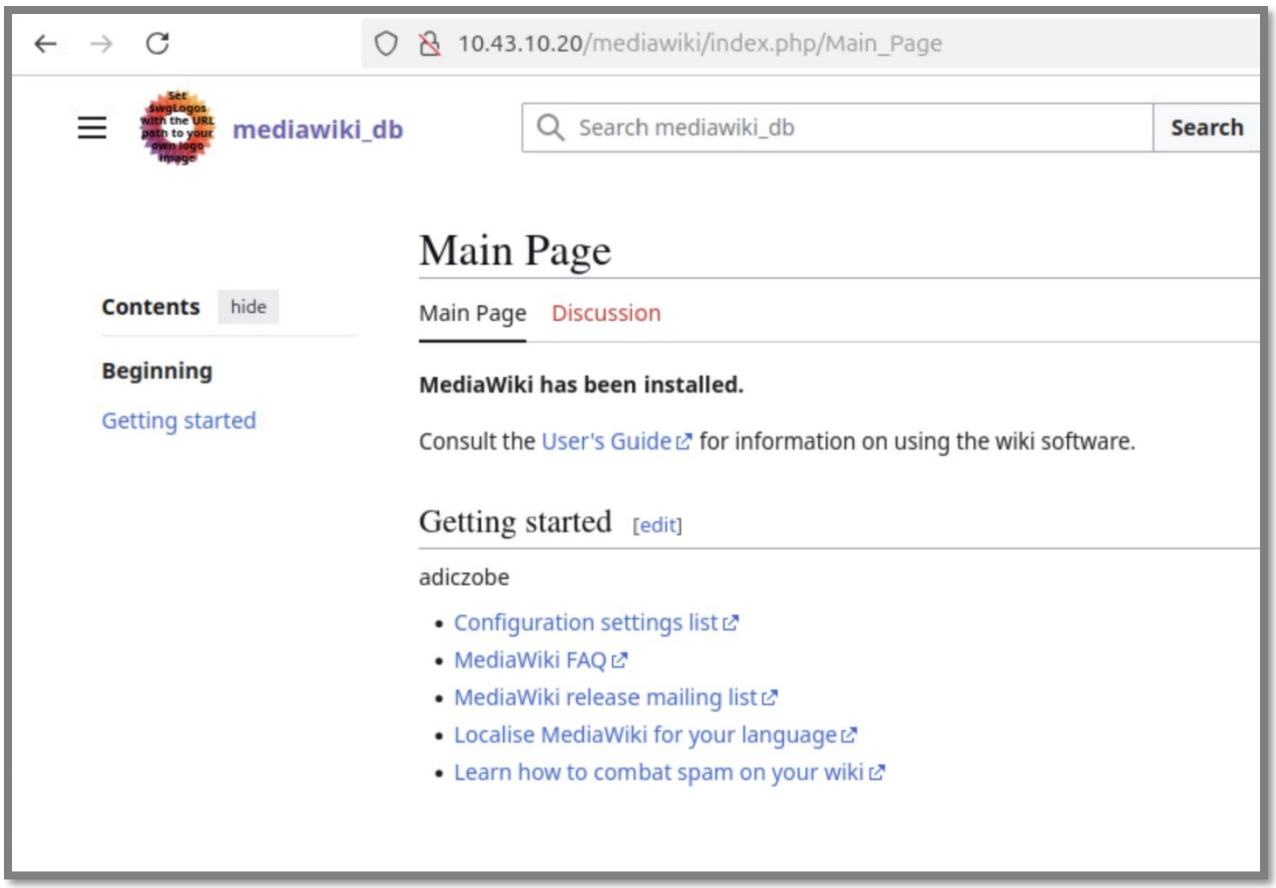


Figure 5: Main Page with UBIT

The main page of MediaWiki, accessible on **UbuntuClient**, I edited it by modifying the "Getting Started" section. My UBIT name was added to the main page as part of this edit.

4. Network Firewall Hardening

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✗	0/0 B	IPv4 *	OPT1 subnets	*	LAN subnets	*	*	none		Block Incoming from ServerNet	
✓	0/0 B	IPv4 TCP	10.42.10.12	*	This Firewall (self)	*	*	none		Only allow Win10 Client	
✗	0/0 B	IPv4 TCP	*	*	This Firewall (self)	*	*	none		Default Block for all devices acessing pfsense	
✓	0/0 B	IPv4 TCP	*	*	10.43.10.20	80 (HTTP)	*	none		Allow HTTP outbound from Adminnet inbound to Servernet	
✓	0/0 B	IPv4 TCP	*	*	10.43.10.20	22 (SSH)	*	none		Allow SSH outbound from Adminnet Inbound to Servernet	
✗	0/0 B	IPv4 TCP	*	*	10.43.10.1/24	*	*	none		Block Everything Except HTTP and SSH	

Figure 6: Firewall Rules on LAN

In order to minimize outbound connectivity from **AdminNet** to **ServerNet**, new firewall rules were added to the **pfSenseRouter**. The first rule allows only **HTTP** and **SSH** connections from **AdminNet** devices to the **UbuntuWebServer**, ensuring that only these protocols are allowed. Additionally, a block rule was created to deny all other traffic, including **Ping**, **HTTPS**, and **DNS**, while still allowing **HTTP** and **SSH**. These rules were integrated into the existing firewall configuration, with some rules rearranged to maintain system integrity. This setup ensures that **AdminNet** devices can only connect to the **UbuntuWebServer** via the specified protocols, enhancing security between the two networks.

Floating WAN LAN OPT1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	IPv4 TCP	*	*	10.43.10.20	80 (HTTP)	*	none		Allow inbound from External Interface	
✓	0/0 B	IPv4 TCP	192.168.13.23	*	10.42.10.12	5985 - 5986	*	none		Allow Winrm inbound from outside device to Win10	
✓	0/0 B	IPv4 TCP	192.168.13.23	*	10.42.10.7	22 (SSH)	*	none		Allow SSH inbound from outside device to Ubuntu	
✓	0/0 B	IPv4 TCP	192.168.13.23	*	10.42.10.12	3389 (MS RDP)	*	none		Allow RDP inbound from outside device to Win10	
✗	0/0 B	IPv4+6 *	*	*	*	*	*	none		Default Block All	

Figure 7: Firewall Rules on WAN

In order to secure the network while allowing access to the MediaWiki webpage, a firewall rule was created on the **WAN** interface. This rule allows inbound connections over the **HTTP** protocol from

any external or web device to access the MediaWiki hosting service. All other types of inbound traffic are blocked. This **pinhole rule** ensures that only HTTP requests can reach the team's architecture, providing controlled access while maintaining security.

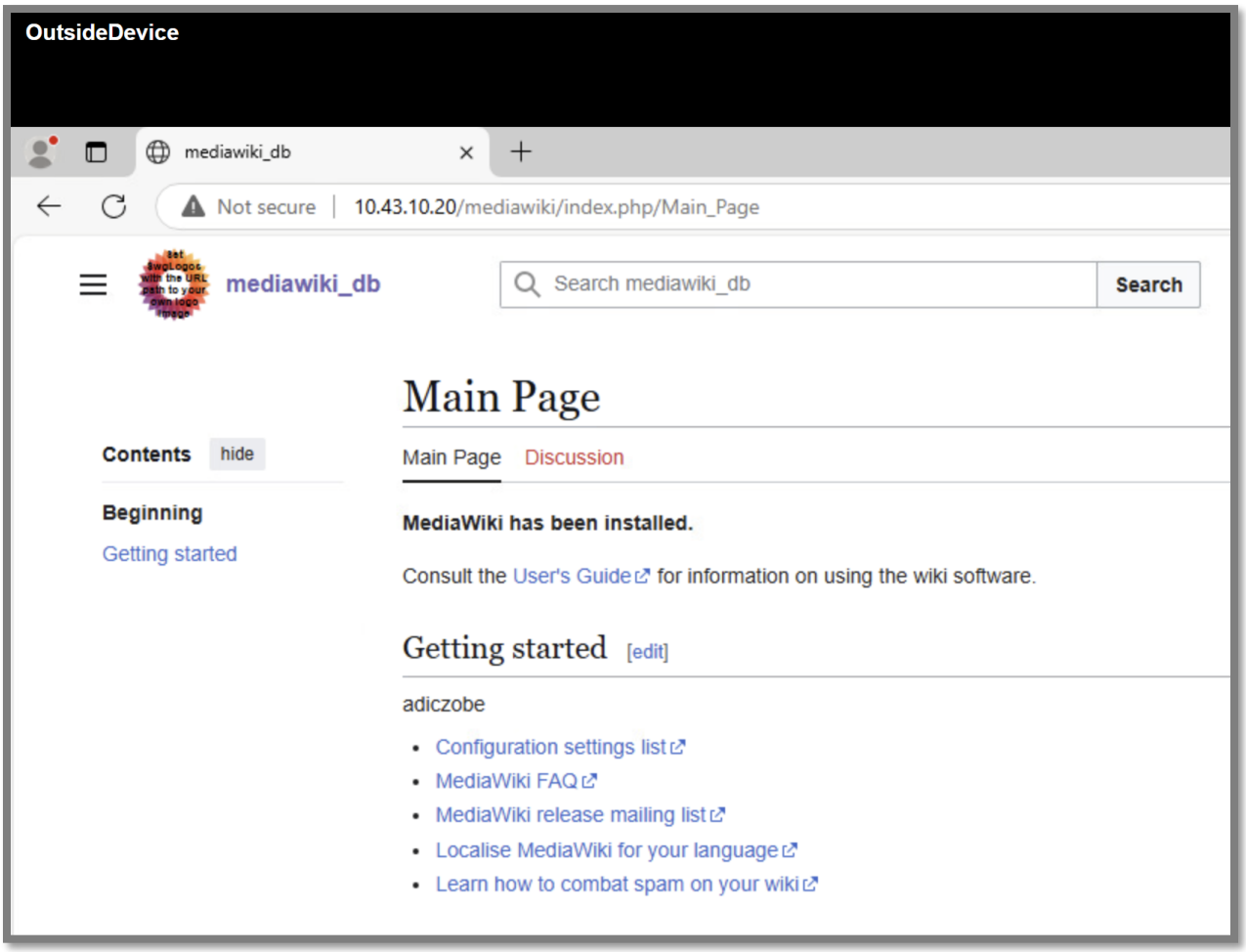


Figure 8: OutsideDevice Can Access the MediaWiki Page

Accessed **OutsideDevice** and navigated to the IP address **10.43.10.20**, which is the **MediaWiki** main page. The connection was successful, shwoing that the **MediaWiki** page is accessible from the external device.

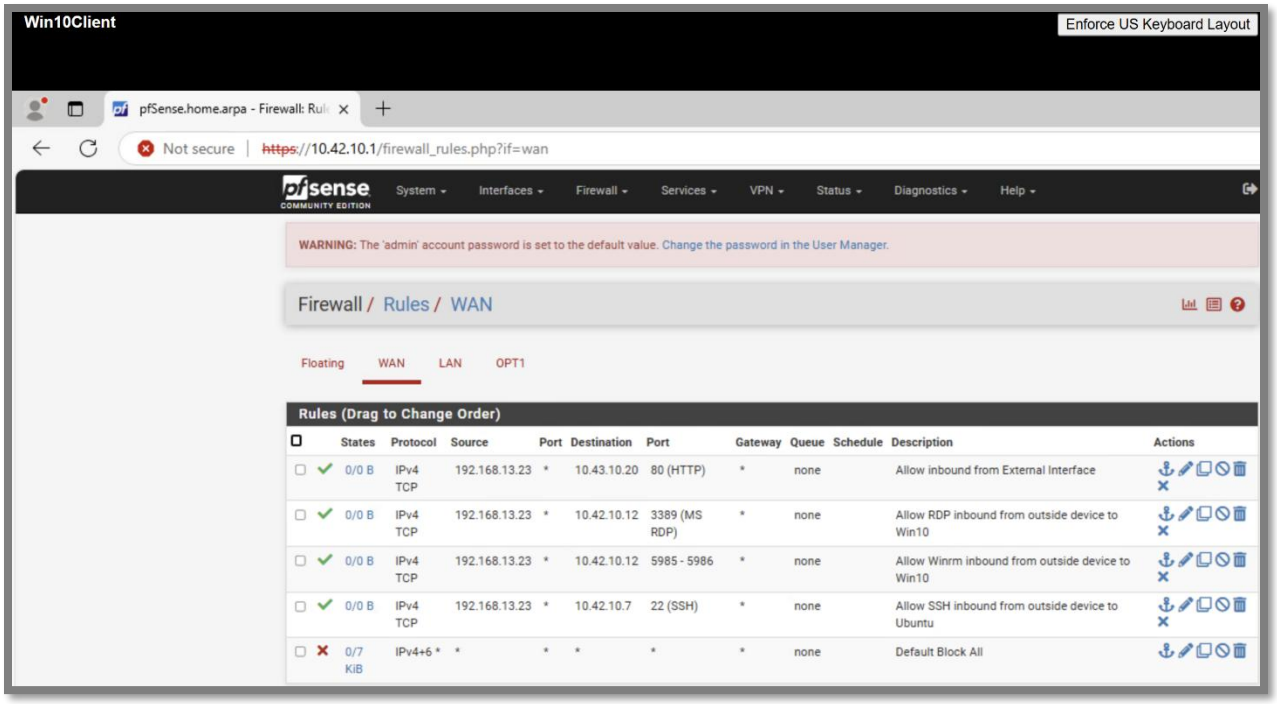


Figure 9: One AdminiNet Device can Access the pfSense webConfigurator GUI

Here you can see **Win10Client** can access the **pfSense webConfigurator GUI**. This is the only device from **AdminNet** permitted to connect to **pfSense**, as all other devices have been blocked by the firewall.

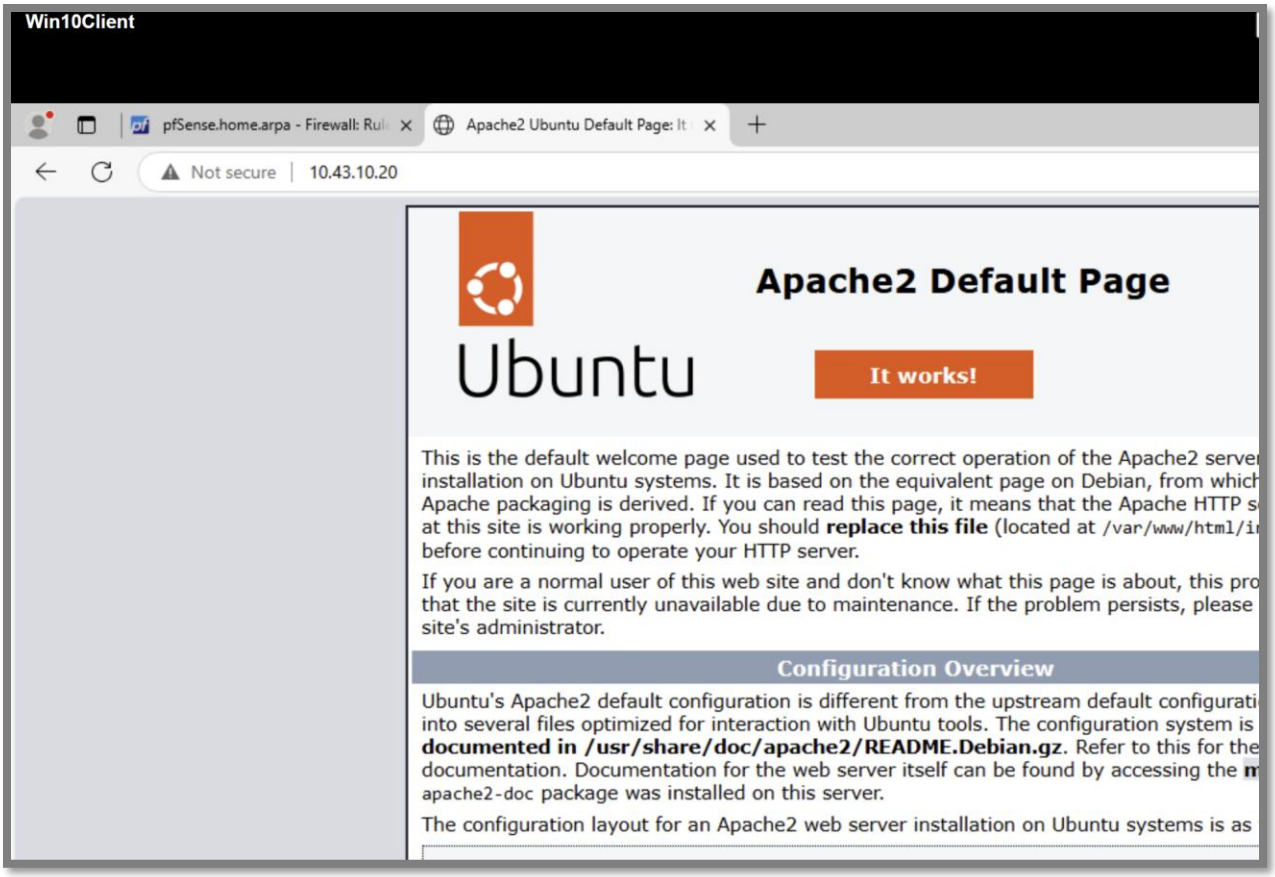


Figure 10: AdminNet Device Can Access UbuntuWebServer via HTTP

Here you can see **Win10Client**, a device from **AdminNet**, accessed **http://10.43.10.20** successfully, confirming that the device can reach the **UbuntuWebServer** via **HTTP**.

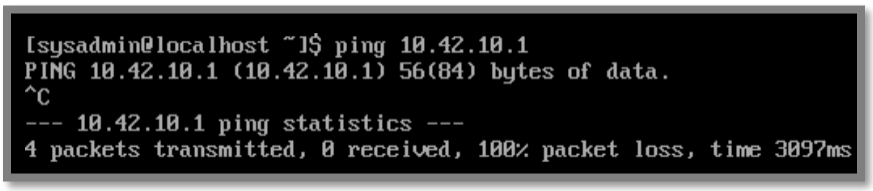


Figure 11: ServerNet Cannot Access AdminNet Devices

I attempted to ping **10.43.10.12** from **RockyDBServer**, but the request was unsuccessful, confirming that **ServerNet** cannot access devices on **AdminNet**.

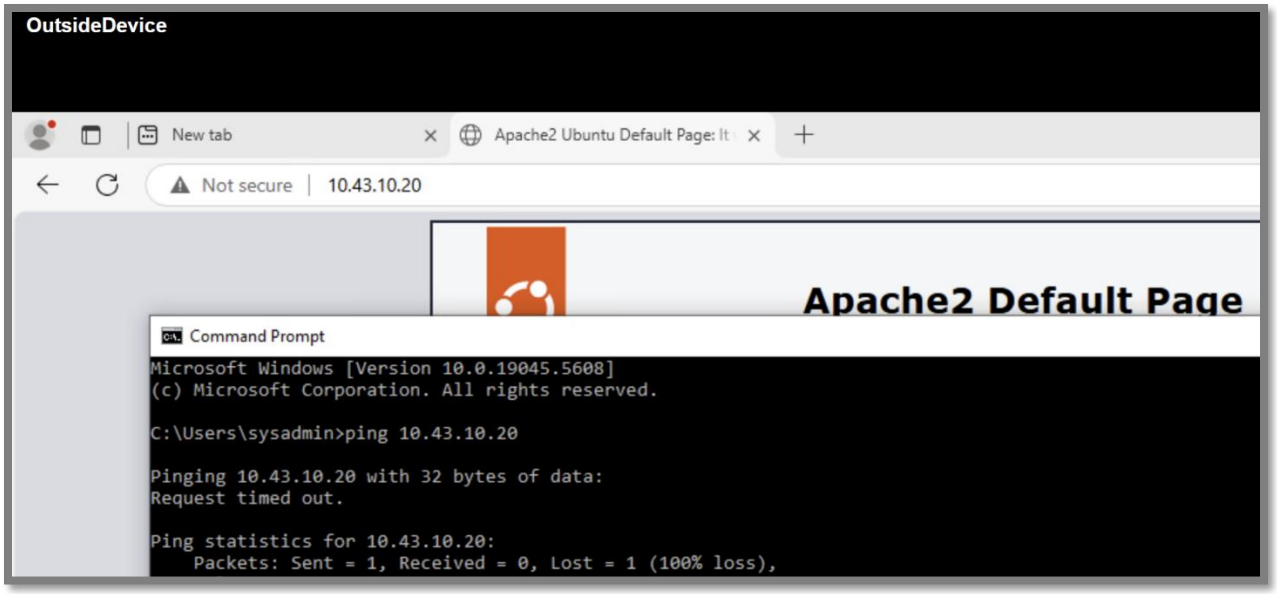


Figure 12: OutsideDevice Cannot Access ServerNet (except for http)

Above I Attempted to ping 10.43.1.1 from **OutsideDevice**, but the request timed out, showing that **OutsideDevice** cannot access **ServerNet**.

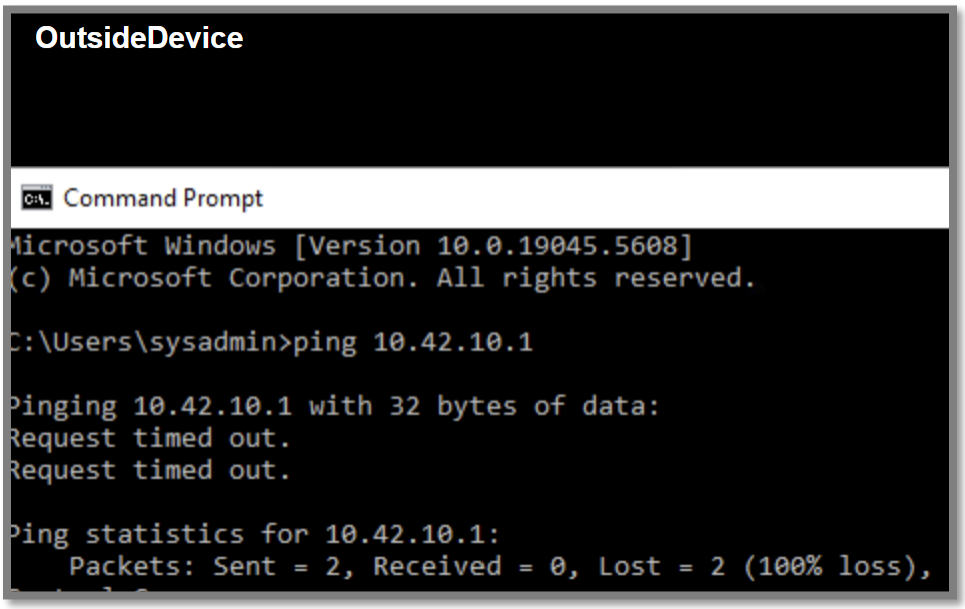


Figure 13: OutsideDevice Cannot Access AdminNet

Attempted to ping 10.43.10.1 from **OutsideDevice**, but the request timed out, indicating that **OutsideDevice** cannot access **AdminNet**.

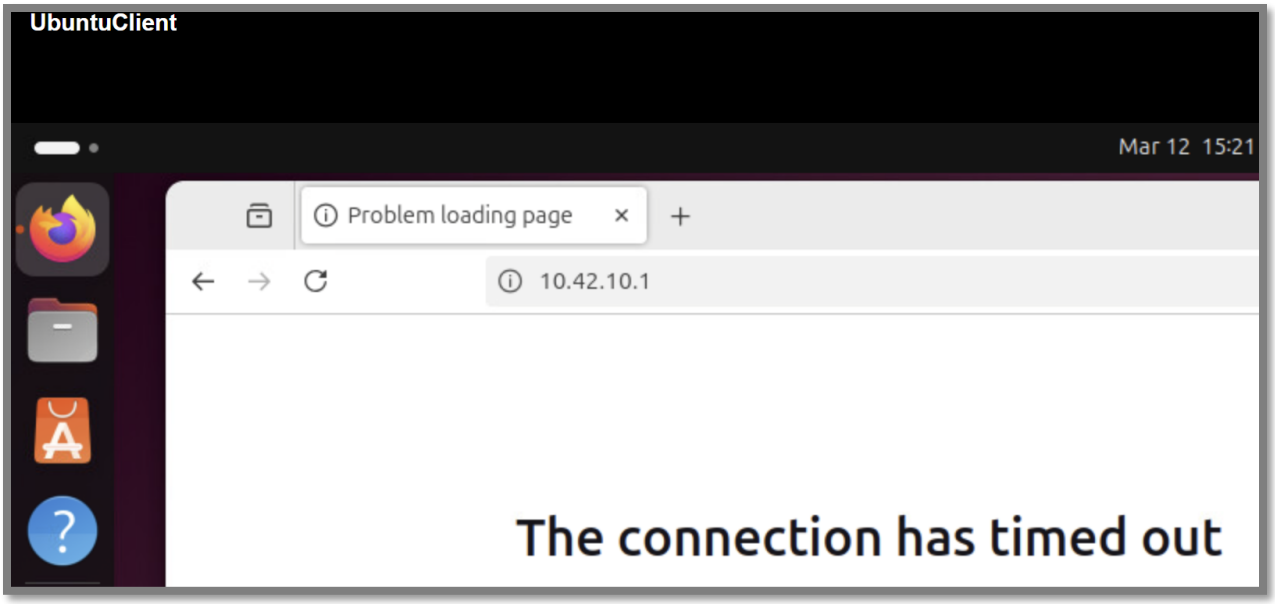


Figure 14: AdminNet Cannot Access pfSense

UbuntuClient, attempted to access **pfSense** using the IP address **10.42.10.1** but was unable to do so. This indicates that **AdminNet** cannot access pfsense

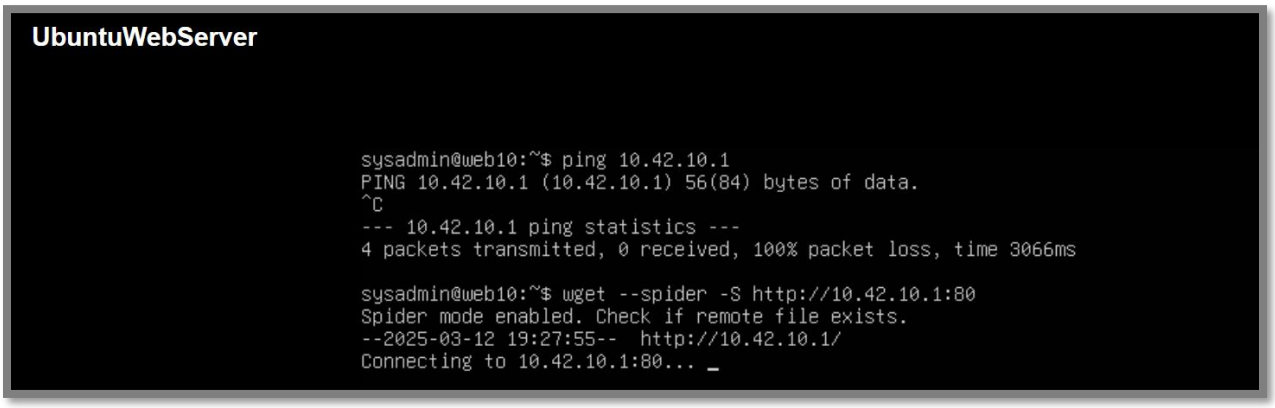


Figure 15: ServerNet Cannot Access pfSense

Attempted to ping **10.42.10.1** from **UbuntuWebServer**, but the response indicated that the destination host is unreachable, confirming that **ServerNet** cannot access **pfSense**.

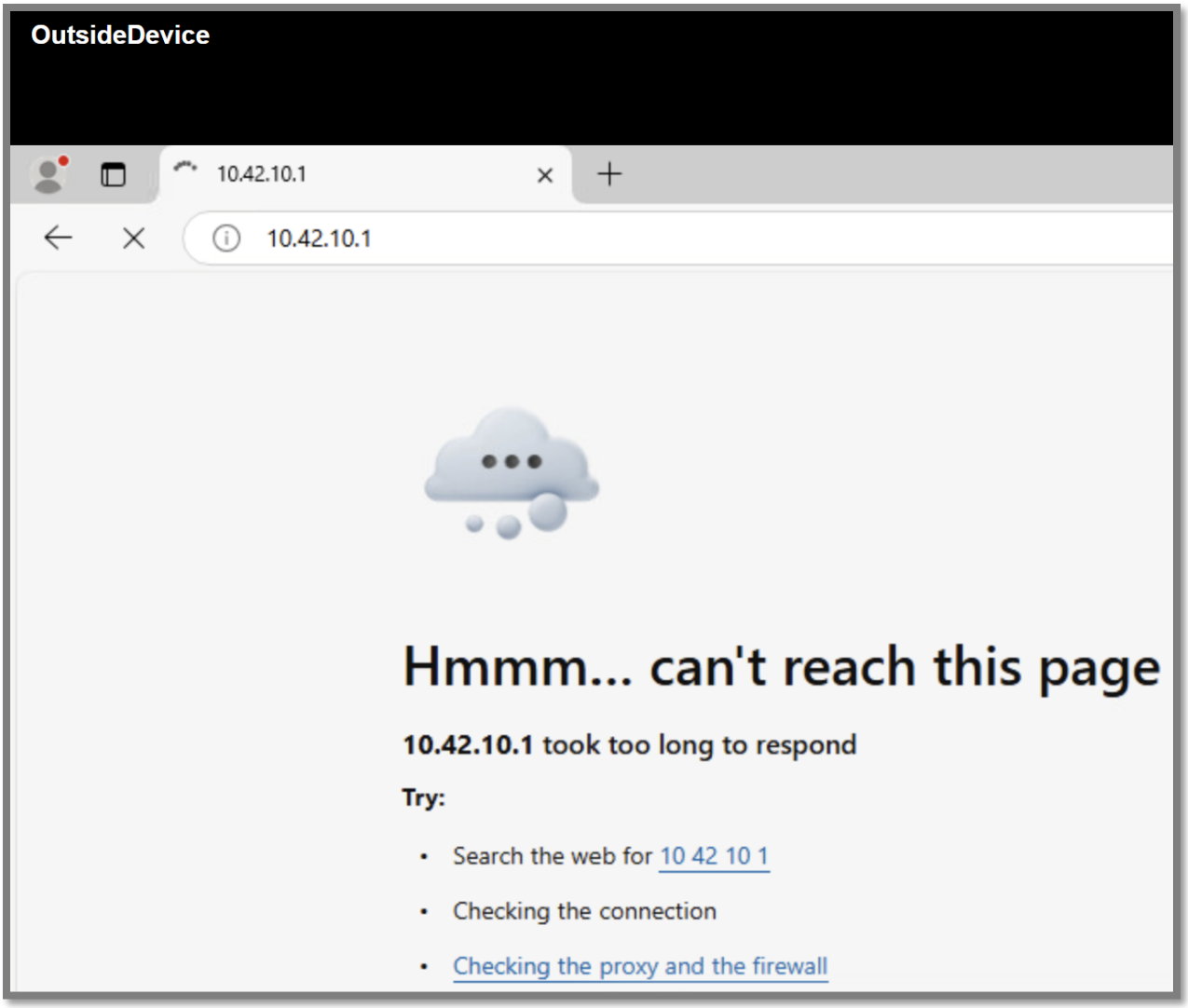


Figure 16: OutsideDevice Cannot Access pfSense

Attempted to access the IP address **10.42.10.1** from **OutsideDevice**, but it was unreachable, indicating that **OutsideDevice** cannot access **pfSense**.

5. SSH Brute Force

Figure 17: Edit File sshd.config

The file **sshd_config** was located on **UbuntuWebServer** and edited as shown in the figure above. After making the necessary changes, the command Control + O was used to write the edits to the file.

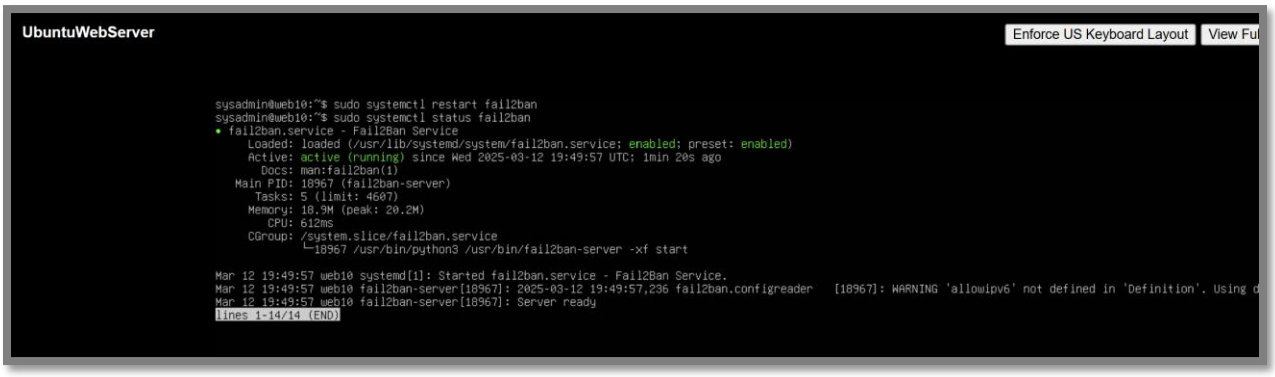
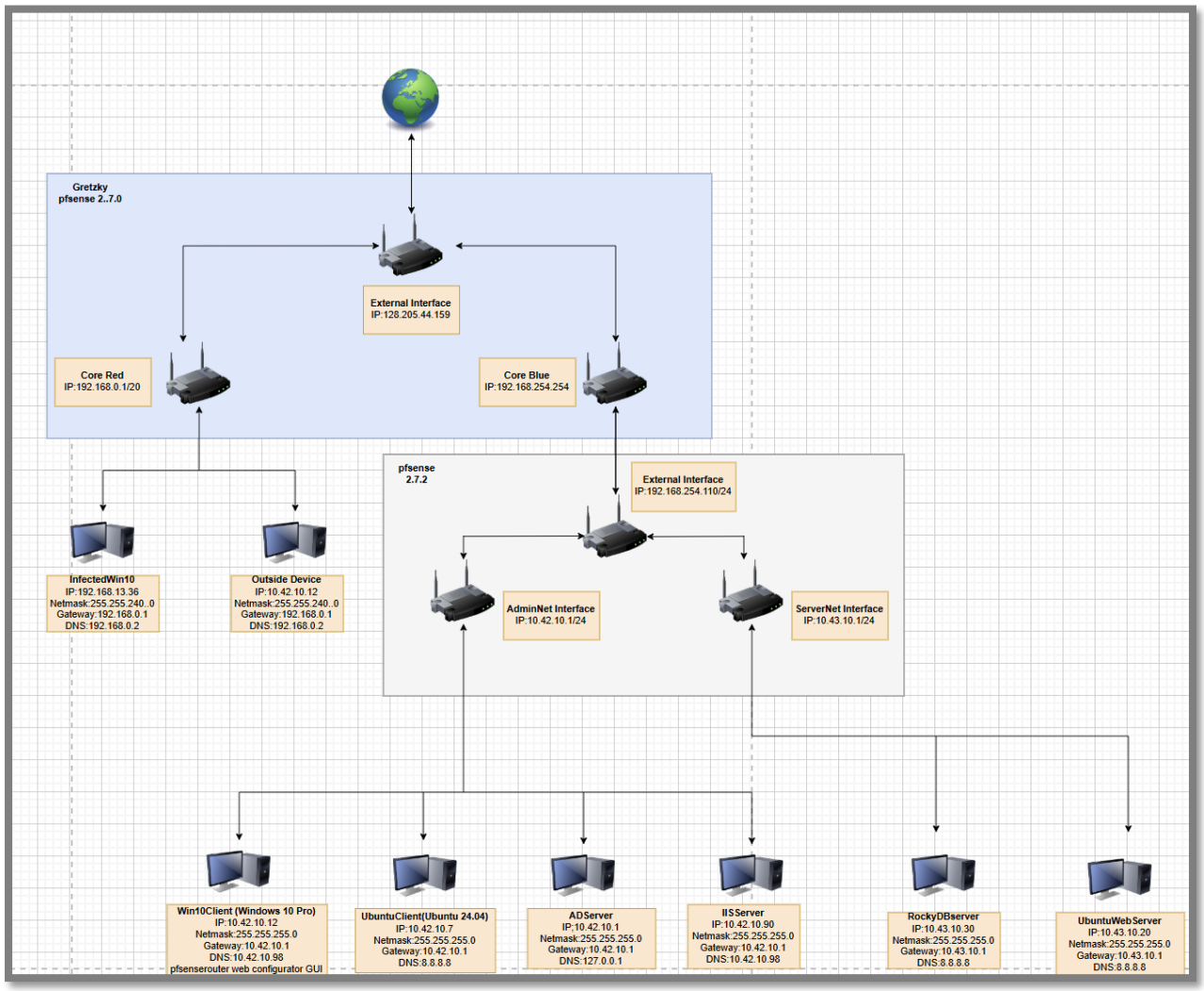


Figure 18: Fail2Ban is Active

The command sudo systemctl restart fail2ban was executed, followed by sudo systemctl status fail2ban. The status indicated that **fail2ban** is active, confirming that it is currently running.

APPENDIX A


Gretzky
pfsense 2..7.0
External Interface
IP:128.205.44.159
Core Red
IP:192.168.0.1/20
Core Blue
IP:192.168.254.254
pfsense
2.7.2
External Interface
IP:192.168.254.110/24
AdminNet Interface
IP:10.42.10.1/24
ServerNet Interface
IP:10.43.10.1/24
InfectedWin10
IP:192.168.13.36
Netmask:255.255.240..0
Gateway:192.168.0.1
DNS:192.168.0.2
Outside Device
IP:10.42.10.12
Netmask:255.255.240..0
Gateway:192.168.0.1
DNS:192.168.0.2
Win10Client (Windows 10 Pro)
IP:10.42.10.12
Netmask:255.255.255.0
Gateway:10.42.10.1
DNS:10.42.10.98
pfsenserouter web configurator GUI
UbuntuClient(Ubuntu 24.04)
IP:10.42.10.7
Netmask:255.255.255.0
Gateway:10.42.10.1
DNS:8.8.8.8
ADServer
IP:10.42.10.1
Netmask:255.255.255.0
Gateway:10.42.10.1
DNS:127.0.0.1
IISServer
IP:10.42.10.90
Netmask:255.255.255.0
Gateway:10.42.10.1
DNS:10.42.10.98
RockyDBserver
IP:10.43.10.30
Netmask:255.255.255.0
Gateway:10.43.10.1
DNS:8.8.8.8
UbuntuWebServer
IP:10.43.10.20
Netmask:255.255.255.0
Gateway:10.43.10.1
DNS:8.8.8.8